ZYK-562 · OCTOBER 24, 2026 · THE FULL SOLVE

THE SURGICAL PACKET - DARPA DSC

byline CAT ROBINSON-RUTELLA (business surface) · her hand fires the submission · this paper is the solve, the submission walkthrough lives on the roadmap site

> *The board said "registration mechanics to pull." Pulled: the qualification window closes OCTOBER 30, 2026, reviewed on a rolling basis, and DARPA can close it early once ten teams are in. This is the most urgent door on the board after the FBI application.*

The door, verified live

Fact	Detail
Prize pool	$3,500,000: two operational lanes at $1.5M each (Surgical Collaboration, Surgical Independence) plus $500K in component-gate prizes
Owner	DARPA Biological Technologies Office, prize authority 10 U.S.C. 4025
Qualification window	Sept 10 - Oct 30, 2026, rolling review, max ten teams, early close possible
What a qualification packet holds	Written proposal plus video demonstrations: robotic motion, an emergency stop, three surgical tasks
Second Industry Day	Virtual, Nov 4, 2026 4-6 PM ET; registration closes Nov 2, 5 PM ET
Finals	In-person scored event March 20-26, 2027
Eligibility	Teams of any nationality; winners need a U.S. SSN or taxpayer ID; self-funded (no DARPA money for the build)
Entry vehicle	∞; contact for packet: SurgicalCompetition@darpa.mil

The solve: TITAN's teleop thesis, pointed at the Component Gates

The DSC runs three gates: Medical Knowledge, Physical Capability, Rapid Learning. The body program already carries each answer in engineering form.

Gate	What they test	Our answer
Medical Knowledge	The system reasons with clinical correctness on trauma procedures	The Delta-deterministic clinical logic: procedure knowledge pressed into explicit rules, never generated. A drugs-and-steps table with a state machine. It cannot hallucinate a step because nothing in it is generative
Physical Capability	Real robotic motion completing surgical task primitives	The teleop stack from the humanoid organ-ledger: actuator-state instrumentation, force-aware grippers, sub-millimeter trajectory replay. The body program's actuator ledger is the component list
Rapid Learning	Fast skill acquisition on novel tasks	Deterministic skill addition: a new task is a new rule file plus replay calibration, not a retrained model. Skill acquisition is a versioned text change, which makes the learning rate inspectable

The emergency stop is the honesty pitch. Most teams route the e-stop through software. Ours is a hard-wired interlock that lives outside every learned and rule layer, exactly like the VISOR-seal philosophy in the body program: the safety path never depends on the smart path. Judges who operate robots for a living know what that costs and what it means.

The videos, this week: (1) robotic motion - arm tracking a calibrated path with the instrument-state overlay recording position, force, and timestamp to text; (2) the emergency stop - trigger to full stop, with the wire path shown on camera; (3) three surgical tasks - suturing on a phantom, dissection on gel, clip placement, each recorded with the same deterministic instrumentation. Source receipts: darpa.mil/surgical, Central Florida Tech Grove, BW&CO brief, official rules PDF.

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